

Theory of Orbital Magnetism of Bloch Electrons: Coulomb Interactions^{*)}

Hidetoshi FUKUYAMA

Department of Physics, Tohoku University, Sendai

(Received May 30, 1970)

Green function formulations of weak field orbital magnetism of Bloch electrons are developed so as to take into account not only interband effects but also mutual interactions between electrons. The modified Bloch functions of Luttinger and Kohn are employed as basic functions to yield gauge- and phase-invariant expressions which, applied to the perfect crystal without many-body interactions, are equivalent to the exact formula. As examples of applications of the formulations for interacting Bloch electron systems, we first discuss the strongly correlated electron system represented by Hubbard's model Hamiltonian within his first approximation. The interband effect between two bands separated off by a Coulomb interaction is shown to have properties similar to Van-Vleck paramagnetism. Next we considered the effect of the correlation in the framework of a random phase approximation in the model of the periodic potential of the one-dimensional weak cosine type.

§ 1. Introduction

Since Landau¹⁾ discussed the orbital magnetism of free electrons, much consideration has been given to the weak magnetism due to the orbital motion of electrons. After Landau, Peierls²⁾ discussed the effect of the periodic potential of the crystal lattice and obtained, among others, a term which expresses the properties of the electrons in a crystal, mainly as the modification of energy dispersion $\epsilon_l(\mathbf{k})$ different from that in free space ($\hbar=1$)

$$\chi_{LP} = \frac{e^2}{48\pi^3 c^2} \sum_l \int d\mathbf{k} \left\{ \frac{\partial^2 \epsilon_l}{\partial k_x^2} \frac{\partial^2 \epsilon_l}{\partial k_y^2} - \left(\frac{\partial^2 \epsilon_l}{\partial k_x \partial k_y} \right)^2 \right\} \frac{\partial}{\partial \epsilon_l} f(\epsilon_l), \quad (1.1)$$

where f is the Fermi distribution function $f(\epsilon_l) = [\exp\{\beta(\epsilon_l - \mu)\} + 1]^{-1}$. This is cited as Landau-Peierls formula and has been frequently used in discussion of low field susceptibility. In the 1950's there appear two main tendencies in the progress of the theory. One is to take into account the effect of many-body interactions and elastic potential of impurities on free electrons. The Coulomb interaction between electrons is discussed by many authors.³⁾ Above all, Kanazawa and Matsudaira⁴⁾ discussed most thoroughly the Green function method in a framework consistent with the high density expansion of correlation energy and obtained the susceptibility χ as

$$\chi/\chi_0 = 1 + \frac{\alpha r_s}{6\pi} \left[\ln r_s + 4 + \ln \frac{\alpha}{2\pi} \right], \quad (1.2)$$

^{*)} This work is based on part of a Ph. D. Thesis submitted to the University of Tokyo, 1970.

where χ_0 is the Landau diamagnetism with the Fermi momentum k_F

$$\chi_0 = -mk_F\mu_B^2/3\pi^2 \quad (1.3)$$

and $r_s = me^2/\alpha k_F$, $\alpha = (4/9\pi)^{1/3}$, $\mu_B = e/2mc$. As regards the electron-phonon interaction, Tani⁵⁾ examined it within the normal processes, and Shiba, Fukuyama and Yamaji⁶⁾ discussed it in greater detail, including Umklapp processes and the case of polarons. These investigations revealed that in metals the correction is of the order of magnitude of s/v_F (s : sound velocity, v_F : Fermi velocity), and therefore very small, while in ionic crystals the effect can be interpreted just as the renormalization of polaron masses. Recently, the effect of the fluctuation of the order parameter in superconductors was examined in classical regions.⁷⁾ On the other hand the effect of the impurity potential was first examined by Dingle,⁸⁾ who showed it to be negligible in metals. If the potential is strong enough to have a bound state and the Fermi level is in such energy regions as in the impurity conduction processes, the orbital magnetism is found to be⁹⁾ quite dissimilar to that in metals.

The other tendency is to discuss the effect of periodic potentials in perfect crystals. The perturbations due to magnetic fields have interband matrix elements, and so the orbital susceptibility is not determined by only one band. This fact has been experimentally realized since the discovery of the anomalous diamagnetism of bismuth by Shönberg and Uddin in 1936.¹⁰⁾ Wilson¹¹⁾ and Adams¹²⁾ have tried to derive an exact expression for the susceptibility, including the above mentioned interband effect. In 1959, Hebborn and Sondheimer¹³⁾ obtained a complete formula and later equivalent results were by different methods.^{14), 15)} Recently, Fukuyama and Kubo¹⁶⁾ gave an interpretation of the diamagnetism of bismuth as the interband effect in the presence of a strong spin-orbit interaction.*) These discussions are based on Bloch's picture of electrons, that is, one electron problem with perfect periodicity of lattice potential.

The aim of the present series of investigations is to develop a theory that can treat the many-body interaction between Bloch electrons and the random arrays of lattice atoms. This article deals with the first kind of problem, taking as examples explicitly both the case of a strongly correlated system represented by Hubbard's model Hamiltonian¹⁷⁾ and that of Bloch electrons in a one-dimensional weak cosine potential with interband coupling and Coulomb correlations.

The first example is of interest as, in the case of vanishing Coulomb repulsion, the electrons are represented by a well-defined band picture. If the finite correlations exist, the band splits into two bands between which the perturbation due to magnetic fields has matrix elements. Such situations are examined in a non-magnetic region within the first approximation of Hubbard.¹⁷⁾

The second example is studied from the following standpoint. The effect

*) Recently Buot (Thesis (1970) University of Oregon) gave a similar interpretation.

of the Coulomb interaction in a high density limit is represented by Eq.(1.2). As shown by Fukuyama and Kubo¹⁸⁾ and Ruvalds,¹⁹⁾ the interband effect plays an important role if the Fermi surface begins to touch the zone boundary, and in such a case the Coulomb interaction between these Bloch electrons is expected to modify Eq.(1.2). We assumed the periodic potential to be a one-dimensional cosine type, which may be too simple compared with real metals, but it may provide rather general features of the effect of the periodic potential near the zone boundary.

To treat these problems the Green function approach is employed. In order to assure the essential properties of the gauge invariance of the result, we adopted the modified Bloch functions of Luttinger and Kohn²⁰⁾ as the basic functions, just as we did in the investigation of the Hall effect.²¹⁾ With this representation the thermodynamic potential is expanded in terms of the vector potential in a system where the effects of electron spins, including the spin-orbit interaction, are completely neglected. In § 2 we give general formulations, whereas in §§ 3 and 4 the Hubbard model and the model of simple periodic potential are discussed, respectively. Section 5 includes a brief summary and discussion.

§ 2. General formulation

Magnetic susceptibility in a weak field per unit volume can be calculated by the formula

$$\chi = - \left[\frac{1}{H} \frac{\partial \Omega}{\partial H} \right]_{H=0}, \quad (2.1)$$

where Ω is the thermodynamic potential of the system with chemical potential μ .

$$-\Omega = kT \ln \text{Tr} \exp[-\beta(\mathcal{H} - \mu N)]. \quad (2.2)$$

As was mentioned in § 1 we study the orbital motion of electron, in a magnetic field exclusively; and then the Hamiltonian can be written as

$$\begin{aligned} \mathcal{H} &= \frac{1}{2m} \sum_i \left(\mathbf{p}_i - \frac{e}{c} \mathbf{A}_i \right)^2 + V_p + \mathcal{H}' \\ &= \frac{1}{2m} \sum_i \mathbf{p}_i^2 + V_p + \mathcal{H}' - \frac{e}{2mc} \sum_i (\mathbf{p}_i \mathbf{A}_i + \mathbf{A}_i \mathbf{p}_i) + \frac{e^2}{2mc^2} \sum_i \mathbf{A}_i^2 \\ &\equiv \mathcal{H}_0 + \mathcal{H}' + \mathcal{H}_1 + \mathcal{H}_2, \end{aligned} \quad (2.3)$$

where $\mathcal{H}'$ represents all of the energy operators other than the kinetic energy and the periodic potential. To avoid the unphysical contributions due to the unbounded character of the coordinate operators, we introduce the vector potential $\mathbf{A}$ first with Fourier component $\mathbf{q}$, as was done by Hebborn et al.,¹⁴⁾

$$\mathbf{A}(\mathbf{r}) = \frac{\mathbf{A}_{\mathbf{q}}}{i} (e^{i\mathbf{q}\mathbf{r}} - e^{-i\mathbf{q}\mathbf{r}}) \quad (2.4)$$

and let $\mathbf{q}=0$ in the final expressions. These procedures make the discussion of the gauge invariance clear. From Eq. (2.1), the second order deviation of $\mathcal{Q}$ due to a magnetic field, defined as $\mathcal{Q}^{(2)}$, is sufficient for the calculation of χ ; and in order to compute $\mathcal{Q}^{(2)}$ we employ Green functions. Expanding the field operators in terms of the modified Bloch function²⁰⁾

$$\begin{aligned}\phi(\mathbf{r}) &= \sum_{l\mathbf{k}} \chi_{l\mathbf{k}} a_{l\mathbf{k}}, \\ \chi_{l\mathbf{k}}(\mathbf{r}) &= e^{i\mathbf{k}\cdot\mathbf{r}} u_{l0}(\mathbf{r}),\end{aligned}\quad (2.5)$$

we have

$$\begin{aligned}\mathcal{H}_0 &= \sum_{l,l'} E(\mathbf{k})_{ll'} a_{l\mathbf{k}}^\dagger a_{l'\mathbf{k}}, \\ E(\mathbf{k})_{ll'} &= \left[\frac{\mathbf{k}^2}{2m} + \varepsilon_l(0) \right] \delta_{ll'} + \mathbf{k} \cdot \mathbf{p}_{ll'}/m\end{aligned}\quad (2.6)$$

and

$$\mathcal{H}_1 = -\frac{e}{mci} \mathbf{A}_{\mathbf{q}} \cdot (\mathbf{j}_{\mathbf{q}} - \mathbf{j}_{-\mathbf{q}}), \quad (2.7)$$

$$\mathcal{H}_2 = -\frac{e^2}{2mc^2} \mathbf{A}_{\mathbf{q}}^2 \cdot (\rho_{2\mathbf{q}} + \rho_{-2\mathbf{q}} - 2\rho_0) \quad (2.8)$$

respectively. Here $\varepsilon_l(0)$ is the band energy of l -th band at $\mathbf{k}=0$; and $\mathbf{p}_{ll'}$, $\mathbf{j}_{\mathbf{q}}$ and $\rho_{\mathbf{q}}$ are given by

$$p_{ll'}^\mu = \frac{(2\pi)^3}{\Omega_a} \int_{\text{Cell}} u_{l0}^* \frac{1}{i} \nabla_\mu u_{l'0} d\mathbf{r}, \quad (2.9)$$

(Ω_a is the volume of a unit cell)

$$j_{\mathbf{q}}^\mu = \sum_{\mathbf{k}, l, l'} (k^\mu \delta_{ll'} + p_{ll'}^\mu) a_{l\mathbf{k}}^\dagger a_{l'\mathbf{k}} = \sum_{l, l'} \gamma(\mathbf{k})_{ll'} a_{l\mathbf{k}}^\dagger a_{l'\mathbf{k}}, \quad (2.10)$$

$$\rho_{\mathbf{q}} = \sum_{\mathbf{k}, l} a_{l\mathbf{k}}^\dagger a_{l\mathbf{k}}, \quad (2.11)$$

$$\mathbf{k}^\pm = \mathbf{k} \pm \frac{\mathbf{q}}{2}.$$

Equation (2.11) is valid for the limit $\mathbf{q} \rightarrow 0$, and this limiting procedure we take before the momentum integration. The one-particle thermal Green function matrix is defined by

$$\begin{aligned}\mathcal{G}_{ll'}(\mathbf{k}, i\varepsilon_n) &= -\frac{1}{\beta} \int_0^\beta d\tau \int_0^\beta d\tau' e^{i\varepsilon_n(\tau-\tau')} \langle T a_{l\mathbf{k}}(\tau) a_{l'\mathbf{k}}^\dagger(\tau') \rangle \\ &= [i\varepsilon_n + \mu - E(\mathbf{k}) - \Sigma(\mathbf{k}, i\varepsilon_n)]_{ll'}^{-1},\end{aligned}\quad (2.12)$$

where $\varepsilon_n = (2n+1)\pi T$ and Σ is the self-energy matrix. With these notations, $\mathcal{Q}^{(2)}$

of the system with translational symmetry can be written as (with spin multiplicity 2)

$$\begin{aligned} \Omega^{(2)} = & \left(\frac{e}{mc} \right)^2 \sum_{\mu, \nu} A_{\mathbf{q}\mu} A_{\mathbf{q}\nu} T \sum_n \sum_{\mathbf{k}} \text{Tr} [\gamma_\mu \mathcal{G}(\mathbf{k}^+, i\varepsilon_n) \Gamma_\nu(\mathbf{k}^+, \mathbf{k}^-; i\varepsilon_n) \mathcal{G}(\mathbf{k}^-, i\varepsilon_n) \\ & + \gamma_\nu \mathcal{G}(\mathbf{k}^+, i\varepsilon_n) \Gamma_\mu(\mathbf{k}^+, \mathbf{k}^-; i\varepsilon_n) \mathcal{G}(\mathbf{k}^-, i\varepsilon_n)] + 2 \frac{e^2}{mc^2} A_{\mathbf{q}}^2 T \sum_n \sum_{\mathbf{k}} \mathcal{G}(\mathbf{k}, i\varepsilon_n), \end{aligned} \quad (2.13)$$

where $(e/m)\Gamma_\mu(\mathbf{k}^+, \mathbf{k}^-, i\varepsilon_n)$ is the current vertex function between two states $(\mathbf{k}^+, i\varepsilon_n)$ and $(\mathbf{k}^-, i\varepsilon_n)$. Tr is meant to take trace over band indices; and momentum summation over $\mathbf{k}$ must be performed within the first Brillouin zone, such that $\mathbf{k}^\pm$ are also in this zone. The contribution (2.13) is shown diagrammatically in Fig. 1. If a magnetic field lies in z -direction, $\Omega^{(2)}$ should contain a factor

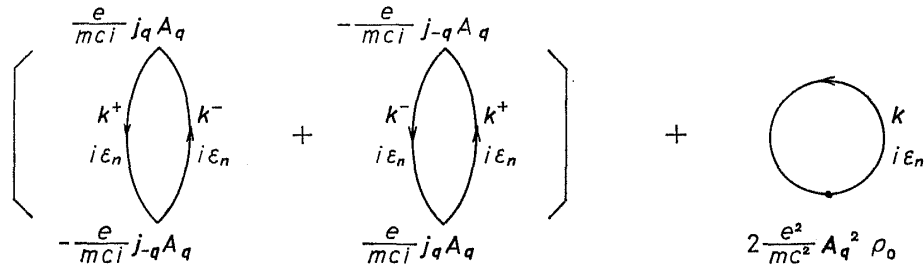


Fig. 1. Contribution to thermodynamic potential in the second order of a magnetic field.

$(q_x A_{qy} - q_y A_{qx})^2$; and for such a term to appear, the self-consistency of the approximations for the self-energy function Σ and the vertex corrections for Γ_μ is needed. If the interaction is such that the momentum dependence of Σ is negligible and so the vertex corrections are not important, we can replace Γ_μ by γ_μ . This approximation is, of course, exact for the perfect crystal without $\mathcal{H}'$. In this case (2.13) can be written in the limit of small $\mathbf{q}$ (Appendix A)

$$\Omega^{(2)} = -4 \frac{\mu_B^2}{m^2} (q_x A_{qy} - q_y A_{qx})^2 T \sum_n \sum_{\mathbf{k}} \text{Tr} \mathcal{G} \gamma_x \mathcal{G} \gamma_y \mathcal{G} \gamma_x \mathcal{G} \gamma_y. \quad (2.14)$$

Correspondingly, the susceptibility, defined by Eq. (2.1) or equivalently by

$$-\Omega^{(2)} = \frac{1}{2} \chi \overline{\mathbf{H}(\mathbf{r})^2} = \chi \cdot (\mathbf{q} \times \mathbf{A}_{\mathbf{q}})^2$$

is given by

$$\chi = 4 \frac{\mu_B^2}{m^2} T \sum_n \sum_{\mathbf{k}} \text{Tr} \mathcal{G} \gamma_x \mathcal{G} \gamma_y \mathcal{G} \gamma_x \mathcal{G} \gamma_y. \quad (2.15a)$$

This is obviously invariant under the change of the phase of Bloch functions. In Eq. (2.15a) γ_μ 's and $\mathcal{G}$ are represented by Luttinger-Kohn functions, but they can be considered like those represented by the usual Bloch functions because

these two are related by a unitary transformation. Thus we, hereafter, may consider γ_μ and $\mathcal{G}$ as being those represented by Bloch functions, writing j_μ and G instead of these. That is

$$\chi = 4 \frac{\mu_B^2}{m^2} T \sum_n \sum_k \text{Tr } G j_x G j_y G j_x G j_y. \quad (2.15b)$$

Then, if $\mathcal{H}' = 0$, G is diagonal with respect to band indices

$$G_{ll'} = \delta_{ll'} (i\varepsilon_n + \mu - \varepsilon_l(\mathbf{k}))^{-1} \equiv \delta_{ll'} g_l, \quad (2.16)$$

where $\varepsilon_l(\mathbf{k})$ is the energy of l -th band. Substitution of Eq. (2.16) into (2.15b) yields the exact formula obtained by the preceding authors,^{(13),(14),(15)} and this is proved in Appendix B in connection with the work of Ichimaru.⁽¹⁵⁾ If we take only the intraband matrix elements of $\mathbf{j}$ in Eq. (2.15b), there follows

$$\bar{\chi} = 4m^2 \mu_B^2 T \sum_n \sum_{k,l} g_l^4 \left(\frac{\partial \varepsilon_l}{\partial k_x} \right)^2 \left(\frac{\partial \varepsilon_l}{\partial k_y} \right)^2 \quad (2.17)$$

as $(j_\mu/m)_{ll} = (\partial/\partial k_\mu) \varepsilon_l$. The summation over n gives

$$\begin{aligned} \bar{\chi} = & \frac{m^2 \mu_B^2}{12\pi^3} \sum_l \int d\mathbf{k} \left[\frac{\partial^2 \varepsilon_l}{\partial k_x^2} \frac{\partial^2 \varepsilon_l}{\partial k_y^2} + 2 \left(\frac{\partial^2 \varepsilon_l}{\partial k_x \partial k_y} \right)^2 \right. \\ & \left. + \frac{3}{2} \left(\frac{\partial \varepsilon_l}{\partial k_x} \frac{\partial^3 \varepsilon_l}{\partial k_x \partial k_y^2} + \frac{\partial \varepsilon_l}{\partial k_y} \frac{\partial^3 \varepsilon_l}{\partial k_y \partial k_x^2} \right) \right] \frac{\partial f(\varepsilon_l)}{\partial \varepsilon_l}. \end{aligned} \quad (2.18)$$

This contrasts with the Landau-Peierls (L.P.) approximation (1.1), which can be derived if we adopt a one-band approximation for γ_μ and $\mathcal{G}$ in Eq. (2.13) from the first and then expand these quantities in terms of $\mathbf{q}$. This is the spirit of L.P. approximation. On the other hand, Hebborn, Sondheimer, Luttinger and Stiles⁽¹⁴⁾ give the one-band formula

$$\begin{aligned} \chi_{\text{HS}} = & \frac{m^2 \mu_B^2}{12\pi^3} \sum_l \int d\mathbf{k} \left[\frac{\partial^2 \varepsilon_l}{\partial k_x^2} \frac{\partial^2 \varepsilon_l}{\partial k_y^2} - \left(\frac{\partial^2 \varepsilon_l}{\partial k_x \partial k_y} \right)^2 \right. \\ & \left. + \frac{3}{2} \left(\frac{\partial \varepsilon_l}{\partial k_x} \frac{\partial^3 \varepsilon_l}{\partial k_x \partial k_y^2} + \frac{\partial \varepsilon_l}{\partial k_y} \frac{\partial^3 \varepsilon_l}{\partial k_y \partial k_x^2} \right) \right] \frac{\partial f(\varepsilon_l)}{\partial \varepsilon_l}. \end{aligned} \quad (2.19)$$

Such ambiguity in one-band approximation arises from the fact that the results depend on the stages where we adopt the one-band approximation. In fact, part of the interband matrix elements of j_μ in Eq. (2.15b) can also be written as the intraband contributions by the relation⁽²²⁾

$$\sum_{\nu} \frac{j_{\nu l}^\alpha j_{l\nu}^\beta}{\varepsilon_l - \varepsilon_\nu} = \frac{m}{2} \left(m \frac{\partial^2 \varepsilon_l}{\partial k_\alpha \partial k_\beta} - \delta_{\alpha\beta} \right). \quad (2.20)$$

These three different expressions, which include only band energy pertinent to the problem, are compared with one another in Appendix C for the case of a weak periodic potential which can be treated by perturbation. At any rate, interband

effects are now known^{16),18),19)} to be essential at least for electrons with a small band gap, and so we shall not be bothered about this problem any more in this article.

§ 3. Hubbard model

As an example which can be treated by the formula (2.15) we next treat the system of strongly correlated electrons represented by the Hubbard model¹⁷⁾

$$\mathcal{H} = \sum_{i,j} t_{ij} a_i^\dagger a_j + I \sum_i n_{i\sigma} n_{i-\sigma}. \quad (3.1)$$

One of the main aspects of this Hamiltonian is the discussions about the occurrence of ferromagnetism. If the Hartree-Fock approximation is taken, ferromagnetism occurs, roughly speaking, when I is greater than the band width; but Hubbard showed that the correlation is very unfavorable for this occurrence. In order to discuss weak magnetism, we should treat the non-magnetic case, which is the case in the following discussion. Moreover, the orbital degeneracy, which introduces important effects on orbital magnetism in some cases, is neglected in this model Hamiltonian equation (3.1).

Although the essential features of the dynamical character of the mutual interactions should express themselves in the vertex corrections of the current operator, we confine our present discussion within the first approximation of Hubbard, in which the vertex correction is negligible. Such corrections are expected to be less important for the susceptibility than for the electrical conductivity by the difference of analyticity of the propagators in these two cases. As the Hamiltonian equation (3.1) assumes that there is only one band in the energy range of interest and that many other bands are energetically far apart from this band, we may safely adopt a one-band approximation in Eq. (2.15b). In Eq. (3.1), this corresponds to the approximation which takes into account the magnetic field mostly in the form of modification of transfer energy through the Peierls phase factor.²⁾ This is expected to be valid at least for metals. Thus we have

$$\begin{aligned} \chi &= 4m^2 \mu_B^2 T \sum_n \sum_{\mathbf{k}} G^4(\mathbf{k}, i\varepsilon_n) \left(\frac{\partial \varepsilon}{\partial k_x} \right)^2 \left(\frac{\partial \varepsilon}{\partial k_y} \right)^2 \\ &= -\frac{4m^2 \mu_B^2}{\pi} \text{Im} \sum_{\mathbf{k}} \int dx f(x) G^R(\mathbf{k}, x)^4 \left(\frac{\partial \varepsilon}{\partial k_x} \right)^2 \left(\frac{\partial \varepsilon}{\partial k_y} \right)^2, \end{aligned} \quad (3.2)$$

where

$$\begin{aligned} G^R(\mathbf{k}, x)^{-1} &= x - \varepsilon(\mathbf{k}) - \Sigma(x + i\delta) \\ &\equiv F(x + i\delta) - \varepsilon(\mathbf{k}) \end{aligned} \quad (3.3)$$

and $\varepsilon(\mathbf{k})$ is the Fourier transform of t_{ij} . Hubbard¹⁷⁾ obtained $F(x)$ by decoupling procedures which read as follows:

$$\frac{1}{F(x)} = \frac{1-n/2}{x-\bar{\varepsilon}} + \frac{n/2}{x-\bar{\varepsilon}-I}, \quad (3.4)$$

where $\bar{\varepsilon}$ is the average kinetic energy which we set to be zero and n is the number of electrons per site

$$\begin{aligned} n &= -\frac{2}{\pi} \sum_{\mathbf{k}} \int dx f(x) \operatorname{Im} G^{\mathbf{R}}(\mathbf{k}, x) \\ &\equiv 2 \int dx f(x) \rho(x), \end{aligned} \quad (3.5)$$

where $\rho(x)$ is the “state density” per atom. The poles of $G^{\mathbf{R}}$, or the roots of the equation $F(x) = \varepsilon(\mathbf{k})$, are represented by two real values $x = E(\mathbf{k})_{\pm}$ which corresponds to two bands separated by the Coulomb interaction. By the use of $E_{\pm}(\mathbf{k})$, $G^{\mathbf{R}}$ can be written as

$$G^{\mathbf{R}}(\mathbf{k}, x) = [Z_+(\mathbf{k}) / (x - E_+(\mathbf{k})) + Z_-(\mathbf{k}) / (x - E_-(\mathbf{k}))], \quad (3.6)$$

where

$$Z_{\pm}(\mathbf{k})^{-1} = \frac{\partial}{\partial x} F(x + i\delta) \Big|_{x=E_{\pm}(\mathbf{k})}. \quad (3.7)$$

So far the discussion is independent of lattice structure, but we now explicitly deal with the case of the simple cubic lattice with band energy

$$\varepsilon(\mathbf{k}) = \frac{V_0}{3} (\cos ak_x + \cos ak_y + \cos ak_z), \quad (3.8)$$

where V_0 is half of the band width and a is the atomic spacing. The self-consistent equations (3.4) and (3.5) are complicated even for the band energy (3.8), and so we assume that

$$\rho_0(x) \equiv \frac{1}{N} \sum_{\mathbf{k}} \delta(x - \varepsilon(\mathbf{k})) = \frac{2}{\pi V_0} \left[1 - \left(\frac{x}{V_0} \right)^2 \right]^{1/2}. \quad (3.9)$$

Figure 2 shows the state density curves corresponding to ρ_0 , with exact dispersion (3.8) and the approximation (3.9), respectively. As discussed by Sakurai,²⁸⁾ for electrons with such a density of state the ferromagnetism never occurs even for infinite I within present approximation. In accordance with Eq. (3.9), we adopt the following approximation for Eq. (3.2):

$$\frac{1}{N} \sum_{\mathbf{k}} \left(\frac{\partial \varepsilon}{\partial k_x} \right)^2 \left(\frac{\partial \varepsilon}{\partial k_y} \right)^2 \delta(x - \varepsilon(\mathbf{k})) = \frac{v^4}{30\pi V_0 a^3} \left[1 - \left(\frac{x}{V_0} \right)^2 \right]^{5/2}, \quad (3.10)$$

where v has same dimension as velocity and is of the order of $V_0 a$. The numerical constant in Eq. (3.10) is added for convenience, to be adjusted in the case of small numbers of electrons or holes. Approximation Eq. (3.9) gives Eq. (3.5) in the form

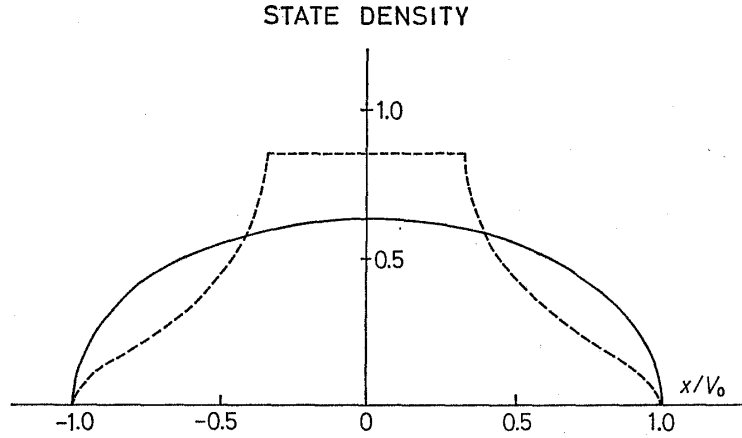


Fig. 2. Exact and approximate curves for state density in unit of V_0 . Broken line is the exact one by Eq. (3.8) and solid line, approximate one by Eq. (3.9).

$$n = \frac{4}{\pi V_0} \int_{-\infty}^{\infty} dx f(x) (1-z^2)^{1/2} \theta(1-z^2), \quad (3.11)$$

where

$$z(x) = F(x + i\delta) / V_0. \quad (3.12)$$

The region where there is a finite density of state is confined within $|z| \leq 1$ and there are two distinct energy regions for x corresponding to two bands $E_{\pm}(\mathbf{k})$. By Eq. (3.10), integration over $\mathbf{k}$ in Eq. (3.2) can be carried out to give

$$\chi = -\frac{m^2 \mu_B^2}{3\pi V_0 a^3} \int dx f(x) \frac{z(3-4z^2)}{\sqrt{1-z^2}} \theta(1-z^2). \quad (3.13)$$

Integration over x in Eqs. (3.11) and (3.12) can be transformed into integration over z . More explicitly, there follows at absolute zero temperature:

I) $n < 1$

$$n = \frac{2}{\pi} \int_{-1}^1 dz \left[1 - \frac{z-s(1-n)}{\{(s-z)^2 + 2zns\}^{1/2}} \right] (1-z^2)^{1/2}, \quad (3.14)$$

$$\chi = -B \int_{-1}^1 dz \left[1 - \frac{z-s(1-n)}{\{(s-z)^2 + 2zns\}^{1/2}} \right] \frac{z(3-4z^2)}{\sqrt{1-z^2}}, \quad (3.15)$$

II) $n > 1$

$$n = \frac{2}{\pi} \int_{-1}^1 dz \left[1 - \frac{z-s(1-n)}{\{(s-z)^2 + 2zns\}^{1/2}} \right] (1-z^2)^{1/2} \\ + \frac{2}{\pi} \int_{-1}^1 dz \left[1 + \frac{z-s(1-n)}{\{(s-z)^2 + 2zns\}^{1/2}} \right] (1-z^2)^{1/2}, \quad (3.16)$$

$$\chi = -B \int_{-1}^1 dz \left[1 - \frac{z-s(1-n)}{\{(s-z)^2 + 2zns\}^{1/2}} \right] \frac{z(3-4z^2)}{\sqrt{1-z^2}}$$

$$-B \int_{-1}^{\nu} dz \left[1 + \frac{z-s(1-n)}{\{(s-z)^2 + 2zns\}^{1/2}} \right] \frac{z(3-4z^2)}{\sqrt{1-z^2}}, \quad (3.17)$$

where $\nu = z(\mu)$, $s = I/V_0$ and $B = \mu_B^2 m^2 v^4 / 6\pi V_0^3 a^3$. Equations (3.14) and (3.16) are self-consistent equations for n with parameter s . We consider n to be an independent variable, and then ν is a function of n . Especially in the case $s=0$, the susceptibility is that of the perfect crystal of a one-electron picture. This is compared with the exact one that is derived from Eqs. (3.2) and (3.8)

$$\begin{aligned} \chi_{\text{EX}} &= -\frac{m^2 \mu_B^2}{12\pi^3} \int d\mathbf{k} \delta(\mu - \epsilon(\mathbf{k})) \frac{\partial^2 \epsilon}{\partial k_x^2} \frac{\partial^2 \epsilon}{\partial k_y^2} \\ &\equiv \chi_0 k_{\text{EX}}(\mu/V_0), \end{aligned}$$

where $\chi_0 = \mu_B^2 m^2 V_0 a / 72\pi$ and

$$k_{\text{EX}}(E) = \int_{-\infty}^{\infty} dt e^{3iEt} J_0(t) J_1^2(t). \quad (3.18)$$

Here J_μ is the Bessel function of the first kind. The approximation Eq. (3.10) corresponds to taking

$$k_{\text{APP}}(E) = (1 - 4E^2)(1 - E^2)^{1/2}. \quad (3.19)$$

Figure 3 shows Eqs. (3.18) and (3.19). Figures 4 and 5 show Φ , defined by $\chi = B\Phi$, for various values of parameter s . In these figures, we also show differential susceptibility $\partial\chi/\partial n$ and the susceptibility $\bar{\chi}$ of “one-band” approximation defined below, respectively. The differential susceptibility per electron in an atomic site is defined as

$$\frac{\partial\chi}{\partial n} = \left(\frac{\partial\chi}{\partial\mu} \right) \left/ \left(\frac{\partial n}{\partial\mu} \right) \right. \equiv 4BD(n) \quad (3.20)$$

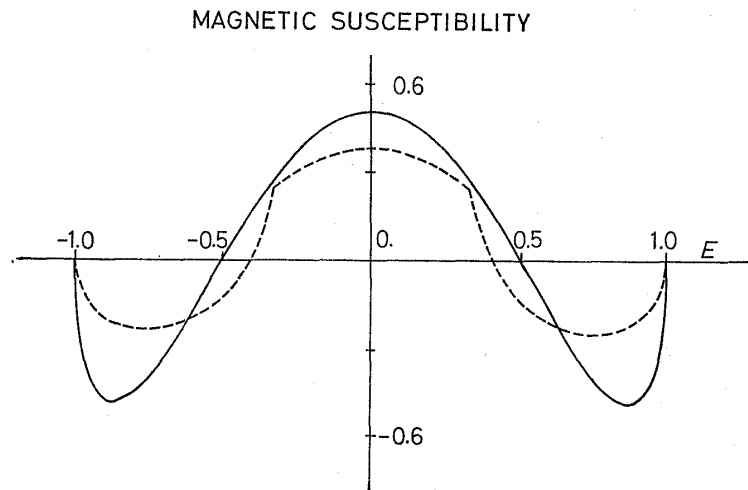


Fig. 3. Comparison of the susceptibilities in the absence of correlations. k_{EX} and k_{APP} are defined by Eqs. (3.18) and (3.19), respectively.

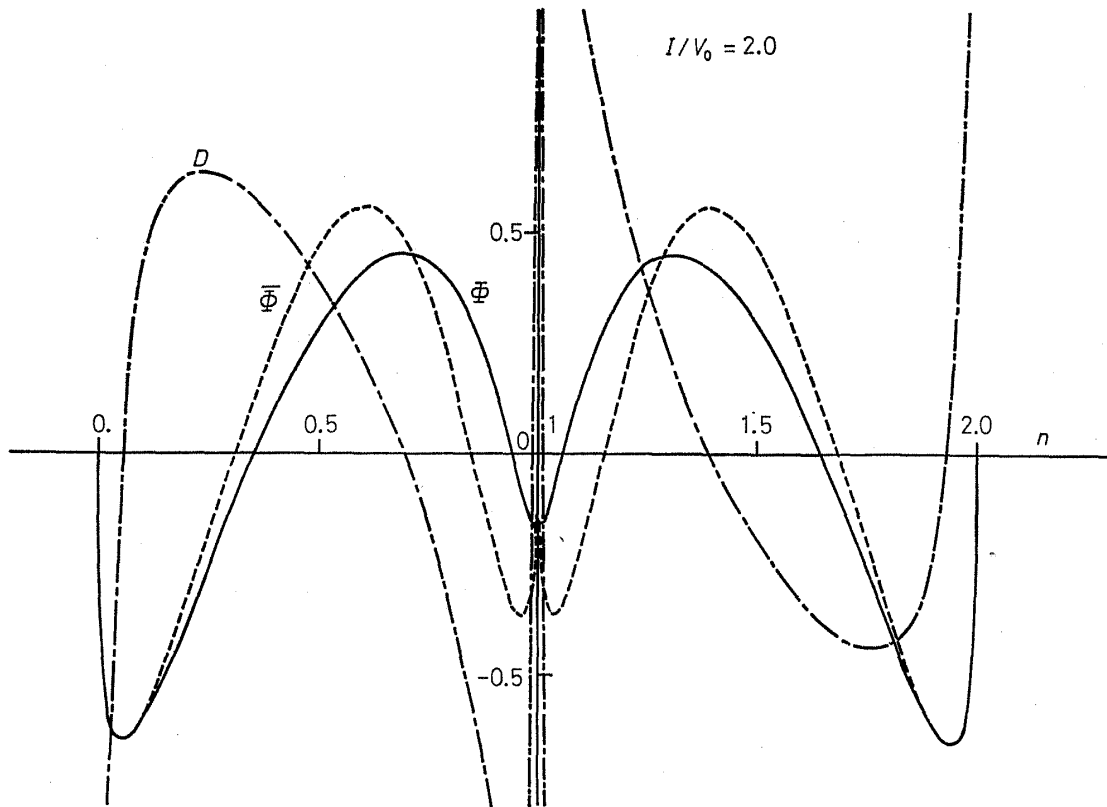


Fig. 4. Susceptibility $\chi/B=\Phi$, differential susceptibility $(\partial\chi/\partial n)/4B=D$ and susceptibility in one-band approximation $\bar{\chi}/B=\bar{\Phi}$ for $s=2$.

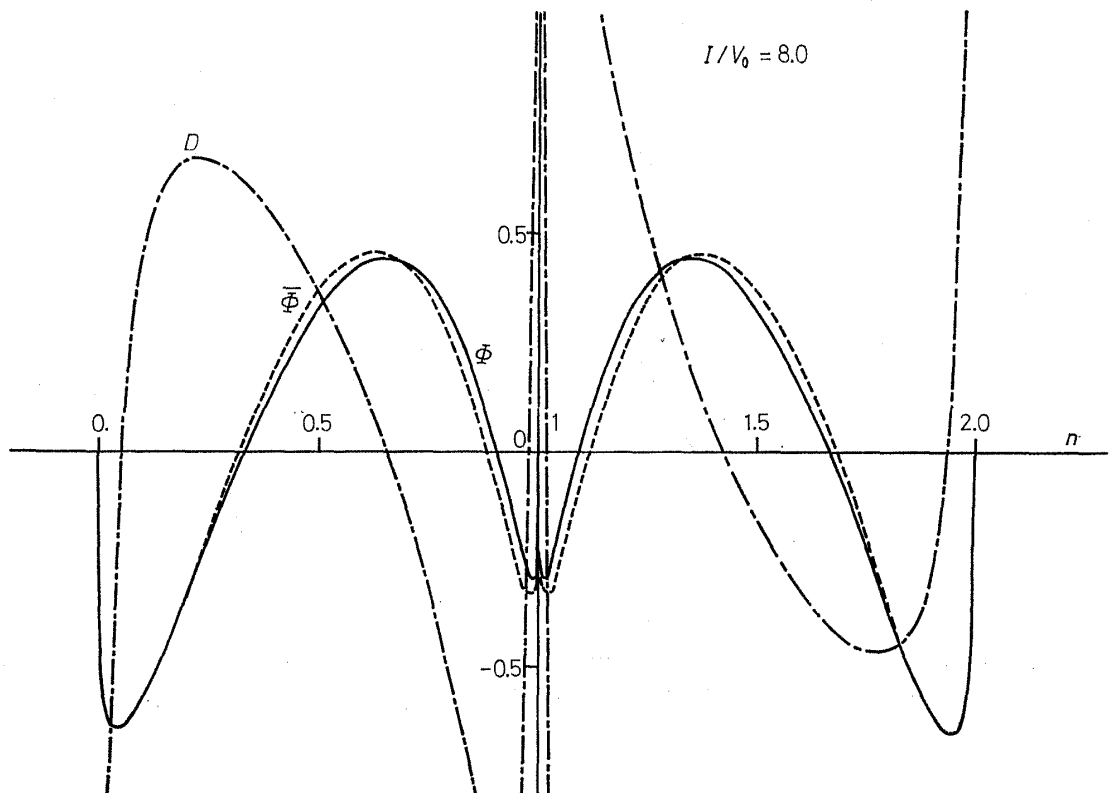


Fig. 5. Φ , D , $\bar{\Phi}$ for $s=8$.

and is expressed by Eqs. (3.14) ~ (3.17) as follows.

I) $n < 1$

$$\frac{\partial \chi}{\partial n} = -\frac{\pi}{2} B[\nu(3-4\nu^2)(1-\nu^2)^{-1}(1-\zeta(\nu, n)) + \eta(\nu, n)], \quad (3.21)$$

II) $n < 1$

$$\frac{\partial \chi}{\partial n} = -\frac{\pi}{2} B[\nu(3-4\nu^2)(1-\nu^2)^{-1}(1-\zeta(1, n) + \zeta(\nu, n)) + \eta(1, n) - \eta(\nu, n)], \quad (3.22)$$

where

$$\zeta(\nu, n) = \frac{2}{\pi} s^2 \int_{-1}^{\nu} dz (1-z^2)^{1/2} [-s+z(1-n)] [(s-z)^2 + 2zsn]^{-3/2}, \quad (3.23)$$

$$\eta(\nu, n) = \frac{2}{\pi} s^2 \int_{-1}^{\nu} dz z (3-4z^2) (1-z^2)^{-1/2} [-s+z(1-n)] \cdot [(s-z)^2 + 2zns]^{-3/2}. \quad (3.24)$$

ζ and η which are shown in Fig. 6 come from the explicit dependence of the one-particle Green function on the number of electrons n , and express the difference between correlated system and rigid band model. In (3.21) and (3.22), $(1-\zeta)$ has effects somewhat like the renormalization constant of a particle state. Due to the relation $\nu(2-n) = -\nu(n)$, the correction term associated with $\nu(3-4\nu^2)(1-\nu^2)^{-1}$ in (3.21) and (3.22) is an odd function of n with respect to $n=1$, whereas the other is even. $\partial\chi/\partial n$ is negative infinite at $n=0$ and $n=1+0^+$ and positive infinite at $n=1-0^+$, and $n=2-0^+$ for finite s , but, if $s=0$, negative infinite at $n=0$ and positive infinite at $n=2$ with a continuous change between these values, as is shown in Fig. 7. This reflects the fact that the two bands separated by correlations, which emerge from one band, appear to be two well-defined Bloch bands with modifications due to the existence of ζ and η . The divergence of $\partial\chi/\partial n$ at the band edge is the characteristic of propagating particles without lifetime broadening. At the other extreme of atomic electrons, this value is constant independent of n . In fact, if there exists a finite imaginary part for Σ , $\partial\chi/\partial n$ remains finite at these edges. Such a change of properties of susceptibility from one similar to the Bloch band to that of atomic (or localized) electrons is investigated in another paper for the model of binary alloys.²⁴⁾

We introduce the susceptibility of “one-band” approximation $\bar{\chi}$ by setting $G^R(\mathbf{k}, x)^4$ in Eq. (3.2) as follows:

$$\begin{aligned} G^R(\mathbf{k}, x)^4 &= [Z_+/(x-E_+(\mathbf{k})) + Z_-/(x-E_-(\mathbf{k}))]^4 \\ &\cong [Z_+^4/(x-E_+(\mathbf{k}))^4 + Z_-^4/(x-E_-(\mathbf{k}))^4], \end{aligned} \quad (3.25)$$

therefore

$$\bar{\chi} = -\frac{2m^2\mu_B^2}{3\pi} \sum_{\pm} \frac{\partial^3}{\partial \mu^3} f(E_{\pm}) \left(\frac{\partial E_{\pm}}{\partial k_x} \right)^2 \left(\frac{\partial E_{\pm}}{\partial k_y} \right)^2. \quad (3.26)$$

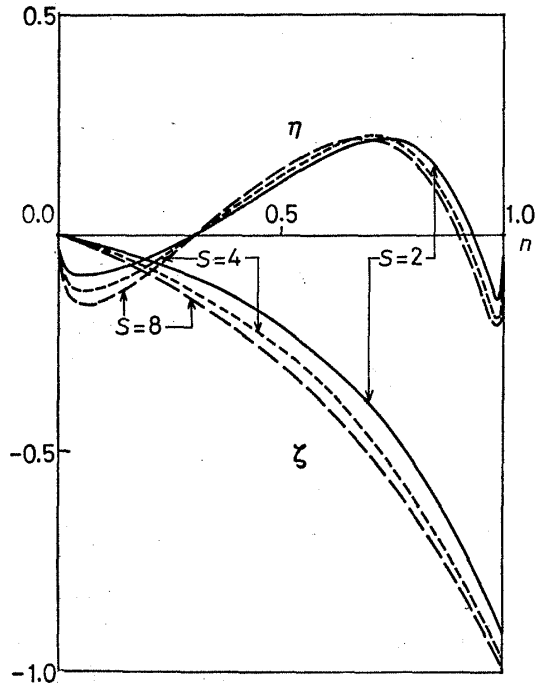


Fig. 6. ζ and η which are defined by Eqs. (3.23) and (3.24) as functions of n .

Here use is made of the relation

$$Z_{\pm} \frac{\partial}{\partial k_{\mu}} \varepsilon = \frac{\partial E_{\pm}}{\partial k_{\mu}}. \quad (3.27)$$

Equation (3.26) expresses the susceptibility of free electrons with band energy $E_{\pm}$. Noting that $E_{\pm} = \mu$, we get $\bar{\Phi} \equiv \bar{\chi}/B$ as follows (Appendix D):

$$\begin{aligned} \bar{\Phi} = \frac{2}{15} \frac{(1-z^2)^{1/2}}{z'^5} [15z^2 z'^4 - 5z'^4(1-z^2) \\ + 25zz'^2 z''(1-z^2) - 3z' z'''(1-z^2)^2 + 12z''^2(1-z^2)^2], \end{aligned} \quad (3.28)$$

where primes on z are meant to take derivative with respect to μ/V_0 . The terms, neglected in Eq. (3.25), have mixing between upper and lower bands, which is automatically included in Eqs. (3.15) and (3.17). From these we define interband contribution $\Delta\chi$ as

$$\Delta\chi = B(\Phi - \bar{\Phi}). \quad (3.29)$$

In the case of a strong correlation, the sign of $\Delta\chi$ can be easily found to be $\Delta\chi < 0$ for $n \cong 0$ and 2 and $\Delta\chi > 0$ for $n \cong 1 \pm 0^+$. These are derived from the fact that the energy operator of a magnetic field has matrix elements between

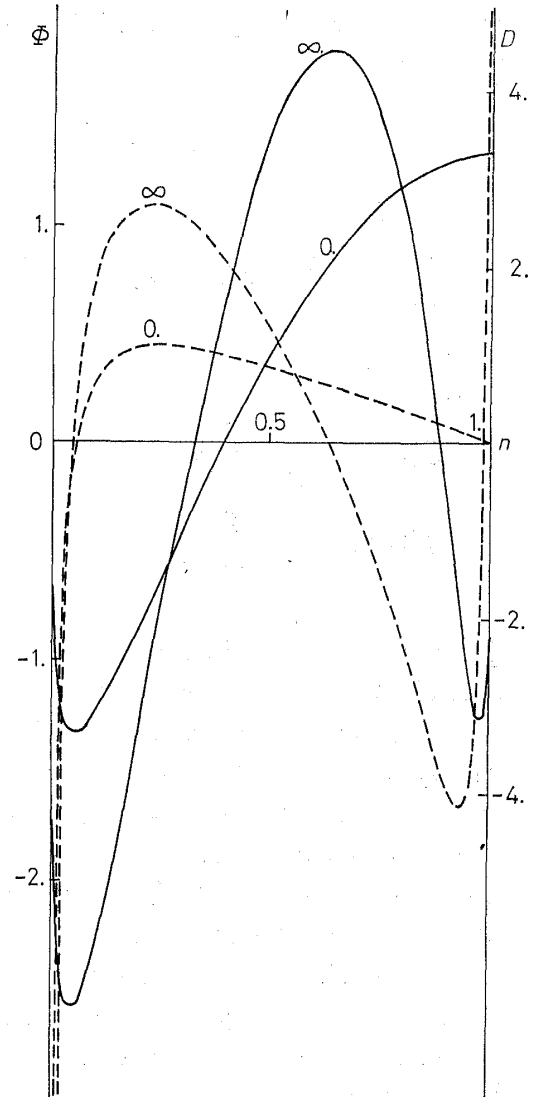


Fig. 7. Φ and D in the limiting cases of $s=0$ and $s=\infty$.

the two states $E_+(\mathbf{k})$ and $E_-(\mathbf{k})$ with same momentum. If $n \cong 0$, the states that contribute to χ lie near the lower edge of the lower band; and these states are mixed together with those in the lower edge of the upper band where the electrons move diamagnetically, as is clear from the curve of $\partial\chi/\partial n$. The situation is the same for other cases for $n > 1$, with a change of sign of energy denominator of the mixing coefficient of wave functions. These features resemble those of Van-Vleck paramagnetism.²⁵⁾

§ 4. Two-band model

In this section we discuss the second problem mentioned in § 1. The Hamiltonian with which we start is ($m = \hbar = 1$)

$$\mathcal{H} = \sum_i \left\{ \frac{1}{2} p_i^2 + 2V \cos \mathbf{K} \mathbf{r}_i \right\} + \sum_{i < j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} \\ \equiv \mathcal{H}_0 + \mathcal{V}, \quad (4.1)$$

where $\mathbf{K} = (0, K, 0)$ is the reciprocal lattice vector, and the magnetic field is assumed to be in the z -direction. Taking $\gamma \equiv V/\epsilon_0 \ll 1$ ($\epsilon_0 = (K/2)^2/2$), we approximate $\mathcal{H}_0$ by two bands pertinent to the problem, retaining the coupling between these bands rigorously. We first expand the field operator by Luttinger-Kohn functions.

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} (\chi_{\mathbf{k}}^{(+)} a_{1\mathbf{k}} + \chi_{\mathbf{k}}^{(-)} a_{2\mathbf{k}}), \quad (4.2)$$

where (volume of crystal is assumed to be a unit)

$$\chi_{\mathbf{k}}^{(+)} = \sqrt{2} e^{i\mathbf{k}\mathbf{r}} \cos(\mathbf{K}\mathbf{r}/2), \\ \chi_{\mathbf{k}}^{(-)} = \sqrt{2} e^{i\mathbf{k}\mathbf{r}} \sin(\mathbf{K}\mathbf{r}/2) \quad (4.3)$$

and then

$$\langle \mathbf{k} | \mathcal{H}_0 | \mathbf{k}' \rangle = \delta_{\mathbf{k}\mathbf{k}'} \begin{pmatrix} \frac{k^2}{2} + V & \lambda k_y \\ \lambda k_y & \frac{k^2}{2} - V \end{pmatrix} \equiv \delta_{\mathbf{k}\mathbf{k}'} E(\mathbf{k}), \quad (4.4)$$

$$\gamma_\mu = \begin{pmatrix} k_\mu & K_\mu \\ K_\mu & k_\mu \end{pmatrix}, \quad (4.5)$$

where $\lambda = K/2$. Equation (4.4) can be diagonalized by a unitary matrix T which depends on the momentum variable such that

$$T(\mathbf{k}) = \frac{1}{\sqrt{2X(X+V)}} \begin{pmatrix} X+V & \lambda k_y \\ -\lambda k_y & X+V \end{pmatrix}, \quad (4.6)$$

where

$$X(\mathbf{k}) = [V^2 + (\lambda k_y)^2]^{1/2} \quad (4.7)$$

and

$$T(\mathbf{k}) E(\mathbf{k}) T^+(\mathbf{k}) = \begin{pmatrix} E_+ & 0 \\ 0 & E_- \end{pmatrix},$$

$$E_{\pm} = \frac{\mathbf{k}^2}{2} \pm X. \quad (4.8)$$

We set up the problem to obtain the leading correction terms to the susceptibility by the Coulomb interaction, which have the same dependence on the expansion parameter r_s as in the case of free electrons, this is, $r_s \ln r_s$. If the Coulomb interaction is absent, Eq. (2.15a) yields, for the present model,

$$\chi^{(0)} = 4\mu_B^2 T \sum_n \sum_{\mathbf{k}} k_x^2 \text{Tr } \mathcal{G}^2 \gamma_y \mathcal{G}^2 \gamma_y. \quad (4.9)$$

Carrying out n -summation and momentum integration we have

$$\begin{aligned} \chi^{(0)} = \chi_K \left[\sum_{\pm} \int_0^1 dy \left(1 \pm \frac{\gamma^2}{(y^2 + \gamma^2)^{3/2}} \right) \theta(\eta_{\pm}) \right. \\ \left. + \frac{3}{2} \gamma^2 \sum_{\pm} \int_0^1 dy \left\{ \frac{\eta_{\pm}}{(y^2 + \gamma^2)^2} \pm \frac{\eta_{\pm}^2}{4(y^2 + \gamma^2)^{5/2}} \right\} \theta(\eta_{\pm}) \right], \end{aligned} \quad (4.10)$$

where $\chi_K = -\mu_B^2 \lambda / 3\pi^2$, $\nu = \mu / \epsilon_0$ and

$$\eta_{\pm} = \nu - y^2 \pm 2(y^2 + \gamma^2)^{1/2}. \quad (4.11)$$

Equation (4.10) is a little different from the results of reference 18), which treated the same model, but which introduced two-band approximation from the first and then expanded in terms of magnetic fields. Nevertheless, these two are numerically almost the same. In order to determine the correction term to Eq. (4.10) of the order of $r_s \ln r_s$, we have to treat rigorously only the long wave part of the Coulomb interaction, as will be shown explicitly in the following; and then we can approximate $\mathcal{V}$ as

$$\mathcal{V} = \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} \frac{4\pi e^2}{q^2} a_{l\mathbf{k}+\mathbf{q}}^{\dagger} a_{l'\mathbf{k}'}^{\dagger} a_{l'\mathbf{k}-\mathbf{q}} a_{l\mathbf{k}}. \quad (4.12)$$

In general, the matrix element of a two-body interaction have four suffices with respect to the band names. This introduces complexity, but extension of the formalism to such a case is straightforward. The change of thermodynamic potential $\Omega^{(2,2)}$ due to $\mathcal{V}$ is shown in Fig. 8. The first two in this figure correspond to the vertex corrections, whereas the others correspond to the corrections of the one-particle propagator. The wavy lines represent the screened Coulomb interaction $v(\mathbf{q}, i\omega_r)$ within a random phase approximation. This approximation for the dielectric function is known to be valid for high density electron gas, and this is also the case for the present model of Bloch electrons, which has a Fermi surface almost similar to that of free electrons. By the procedures to factor out $(q_x A_{qy}$

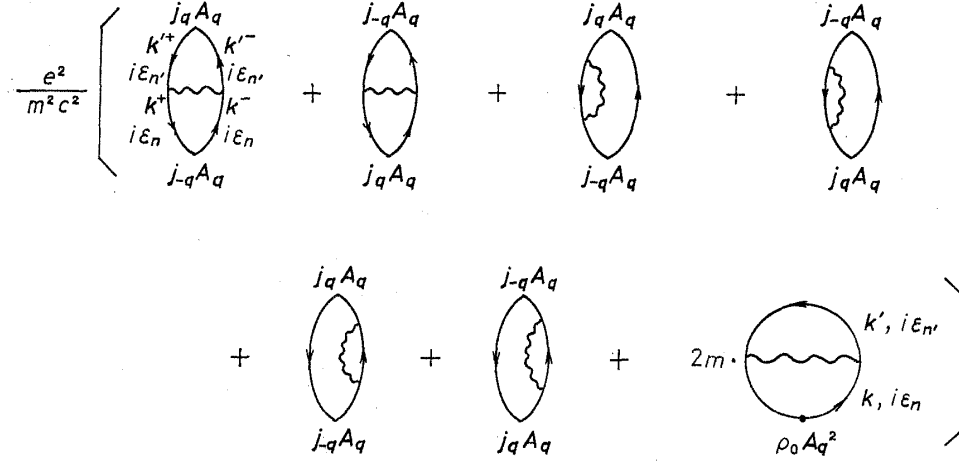


Fig. 8. Contributions to the change of thermodynamic potential due to Coulomb interaction $\mathcal{V}$ in the coefficient of second order of a magnetic field $\mathcal{Q}^{(2,2)}$. Wavy lines represent the screened Coulomb interaction.

$-q_y A_{qx})^2$ as in § 2, we have, after some straightforward calculations, corrections to the susceptibility $\chi^{(2)}$ as follows:

$$\begin{aligned} \chi^{(2)} = & 2\mu_B^2 T \sum_n T \sum_{\mathbf{k}, \mathbf{q}} v(\mathbf{q}, i\omega_n) \\ & \times \text{Tr}[(\mathcal{G}'\gamma'_x \mathcal{G}'\gamma'_y \mathcal{G}' - \mathcal{G}'\gamma'_y \mathcal{G}'\gamma'_x \mathcal{G}') (\mathcal{G}\gamma_x \mathcal{G}\gamma_y \mathcal{G} - \mathcal{G}\gamma_y \mathcal{G}\gamma_x \mathcal{G}) \\ & + 2\mathcal{G}'\{2(\mathcal{G}\gamma_y \mathcal{G}\gamma_y \mathcal{G}\gamma_x \mathcal{G} + \mathcal{G}\gamma_x \mathcal{G}\gamma_x \mathcal{G}\gamma_y \mathcal{G} \\ & - \mathcal{G}\gamma_x \mathcal{G}\gamma_y \mathcal{G}\gamma_y \mathcal{G}\gamma_x \mathcal{G} - \mathcal{G}\gamma_y \mathcal{G}\gamma_x \mathcal{G}\gamma_x \mathcal{G}\gamma_y \mathcal{G}) \\ & + (\mathcal{G}^2\gamma_x \mathcal{G}\gamma_x \mathcal{G} + \mathcal{G}\gamma_x \mathcal{G}\gamma_x \mathcal{G}^2 - \mathcal{G}\gamma_x \mathcal{G}^2\gamma_x \mathcal{G}) \\ & + (\mathcal{G}^2\gamma_y \mathcal{G}\gamma_y \mathcal{G} + \mathcal{G}\gamma_y \mathcal{G}\gamma_y \mathcal{G}^2 - \mathcal{G}\gamma_y \mathcal{G}^2\gamma_y \mathcal{G}) + \mathcal{G}^3\}], \end{aligned} \quad (4.13)$$

where

$$\mathcal{G}' \equiv \mathcal{G}(\mathbf{k} + \mathbf{q}, i\varepsilon_n + i\omega_n) \text{ and } \gamma'_\mu = \gamma_\mu(\mathbf{k} + \mathbf{q}).$$

Here use is made of the approximation of Eq. (4.12) which is exact for $\mathbf{q} \rightarrow 0$ limit. In the derivation of Eq. (4.13), we also made use of the fact

$$\frac{\partial}{\partial k_\mu} \sum_{\mathbf{k}} v(\mathbf{k} - \mathbf{k}', i\omega_n) \mathcal{G}(\mathbf{k}') = \sum_{\mathbf{k}'} v(\mathbf{k} - \mathbf{k}', i\omega_n) \frac{\partial}{\partial k'_\mu} \mathcal{G}(\mathbf{k}'). \quad (4.14)$$

As the effect of retardation is not essential to the problem (see also reference 4) for detailed justifications of the approximation of Fig. 8), we neglect the frequency dependence of $v(\mathbf{q}, i\omega_n)$. The screening constant is, in principle, different from that in a free electron system because of the existence of the periodic potential, which manifest itself in the fact that the polarization diagrams must be represented by a two-by-two Green function matrix, but this modification is small in the present model of a one-dimensional weak cosine potential for which $\gamma \ll 1$. Thus we take

$$v(\mathbf{q}, i\omega_\nu) \cong v(\mathbf{q}) = \frac{4\pi e^2}{q^2 + \kappa^2}, \quad (4.15)$$

$$\kappa^2 = 6\pi n e^2 / \mu, \quad (4.16)$$

where n is the number density of electrons. The summation over n, n' in Eq. (4.13) can be performed if we diagonalize $\mathcal{G}$ by the matrix $T(\mathbf{k})$.

$$\mathcal{G}^B = T \mathcal{G} T^+ = \begin{pmatrix} g_+ & 0 \\ 0 & g_- \end{pmatrix}, \quad (4.17)$$

$$\gamma_\mu^B = T \gamma_\mu T^+ = \begin{pmatrix} (\alpha_+ \mathbf{k})_\mu & -\frac{V}{2X} K_\mu \\ -\frac{V}{2X} K_\mu & (\alpha_- \mathbf{k})_\mu \end{pmatrix}, \quad (4.18)$$

where

$$g_\pm = (i\varepsilon_n + \mu - E_\pm)^{-1} \quad (4.19)$$

and

$$\alpha_{\pm\mu\nu} = \delta_{\mu\nu} \pm \frac{K_\mu K_\nu}{4X}. \quad (4.20)$$

But note that the matrices which diagonalize $\mathcal{G}(\mathbf{k})$ and $\mathcal{G}(\mathbf{k} + \mathbf{q})$ are different because of the difference of momentum. Taking trace over matrix space we have

$$\begin{aligned} \chi^{(2)} = & -\mu_B^2 T \sum_n \sum_{\mathbf{k}} \left[-T \sum_{n'} \sum_{\mathbf{k}'} v(\mathbf{k} - \mathbf{k}') \frac{j_{\text{od}} j_{\text{od}}' k_x k_x' (\delta^2 + \eta^2)}{4XX' (X + V) (X' + V)} (g_+ - g_-) g_+ g_- \right. \\ & \times (g_+' - g_-') g_+' g_-' + 4k_x^2 \left\{ \sum_{\pm} \sigma_{\pm} (g_{\pm}^4 g_{\mp} - g_{\pm}^2 g_{\mp}^3) j_{\text{od}}^2 \right. \\ & + 2(j_+ - j_-) j_{\text{od}} \sigma_{\text{od}} (g_+^2 g_-^3 - g_+^3 g_-^2) \} + 2k_x^2 \sum_{\pm} \sigma_{\pm} g_{\pm}^4 \\ & + 2 \sum_{\pm} \sigma_{\pm} (j_{\pm}^2 g_{\pm}^4 + 2g_{\pm}^3 g_{\mp} j_{\text{od}}^2) \\ & \left. + 2g_+^2 g_-^2 \{ 2\sigma_{\text{od}} j_{\text{od}} (j_+ + j_-) - j_{\text{od}}^2 (\sigma_+ + \sigma_-) \} + 2 \sum_{\pm} \sigma_{\pm} g_{\pm}^3 \right], \quad (4.21) \end{aligned}$$

where

$$\begin{aligned} j_{\pm} &= (\alpha_{\pm} k)_y, \quad j_{\text{od}} = -\frac{VK}{2X}, \\ \sigma_{\pm}(\mathbf{k}) &= \frac{1}{2} \sum_{\mathbf{k}'} v(\mathbf{k} - \mathbf{k}') \left[f(E_+') + f(E_-') \pm \frac{V^2 + \lambda^2 k_y k_y'}{XX'} (f(E_+') - f(E_-')) \right], \\ \sigma_{\text{od}}(\mathbf{k}) &= \frac{1}{2} \sum_{\mathbf{k}'} v(\mathbf{k} - \mathbf{k}') \frac{\lambda(k_y - k_y') V}{XX'} (f(E_+') - f(E_-')), \\ \delta &= \lambda k_y (X' + V) - \lambda k_y' (X + V), \\ \eta &= (X + V) (X' + V) + \lambda^2 k_y k_y'. \end{aligned}$$

Here primes denote that these quantities refer to the momentum $\mathbf{k}'$. The inte-

gration over $\mathbf{k}$ and $\mathbf{k}'$ is exhaustive, but we can determine the coefficient of $r_s \ln r_s$ in Eq. (4.21) as follows. After the summation of n , terms in Eq. (4.21) have the following forms ($f_{\pm}(\mathbf{k}) \equiv f(E_{\pm}(\mathbf{k}))$):

$$\sum_{\mathbf{k}, \mathbf{q}} v(\mathbf{q}) \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} f_{\pm}(\mathbf{k}) f_{\pm}^{(\alpha)}(\mathbf{k}+\mathbf{q}) \equiv \sum_{\mathbf{q}} v(\mathbf{q}) C_{\mathbf{q}}^{(\alpha)}, \quad (4.22)$$

where $\zeta_{\mathbf{k}}$ and $\eta_{\mathbf{k}+\mathbf{q}}$ are analytic functions of the momentum and $f^{(\alpha)}$ is the α -th derivative with respect to the energy. Equation (4.21) includes terms with α up to three. The existence of $r_s \ln r_s$ is due to the divergence of the integration of (4.22) in small $\mathbf{q}$ if $\kappa=0$ (that is, due to whether $C_{\mathbf{q}}^{(\alpha)}$'s contain $1/q$, or not, in the small $\mathbf{q}$ limit). $C_{\mathbf{q}}^{(0)}$ and $C_{\mathbf{q}}^{(1)}$ do not contain such a factor, but $C_{\mathbf{q}}^{(2)}$ and $C_{\mathbf{q}}^{(3)}$ do, as is shown in Appendix E. Thus we get

$$\begin{aligned} \chi^{(2)} &= \alpha r_s \ln r_s \chi_K \Phi, \\ \Phi &= \frac{3}{2\pi} \sum_{\pm} \int_0^1 dy (1 \pm \gamma^2 (y^2 + \gamma^2)^{-3/2}) \phi_{\pm}^{-1} \theta(\eta_{\pm}) \\ &\quad - \frac{1}{\pi} \frac{\partial}{\partial \nu} \sum_{\pm} \int_0^1 dy \left[\pm \frac{\gamma^2 \eta_{\pm}}{(y^2 + \gamma^2)^{3/2}} + \eta_{\pm} + 2 \left(1 \pm \frac{1}{\sqrt{y^2 + \gamma^2}} \right)^2 y^2 \right] \frac{\theta(\eta_{\pm})}{\phi_{\pm}}, \end{aligned} \quad (4.23)$$

where $\eta_{\pm}$ is given by Eq. (4.11) and

$$\phi_{\pm}(y) = \left\{ \nu + \frac{y^2}{y^2 + \gamma^2} \mp \frac{2\gamma^2}{\sqrt{y^2 + \gamma^2}} \right\}^{1/2}. \quad (4.24)$$

Φ is shown graphically in Fig. 9 for $\gamma=0.05$ near the region $\mu=0$. As is expected, Φ has a rather strong dependence on μ . Equation (4.23) reduces $1/6\pi$ for $V=0$, but in the case of $V \neq 0$ there occur discontinuities at $\nu = \pm 1$ on account of large contributions from a small number of electrons near the zone boundary.

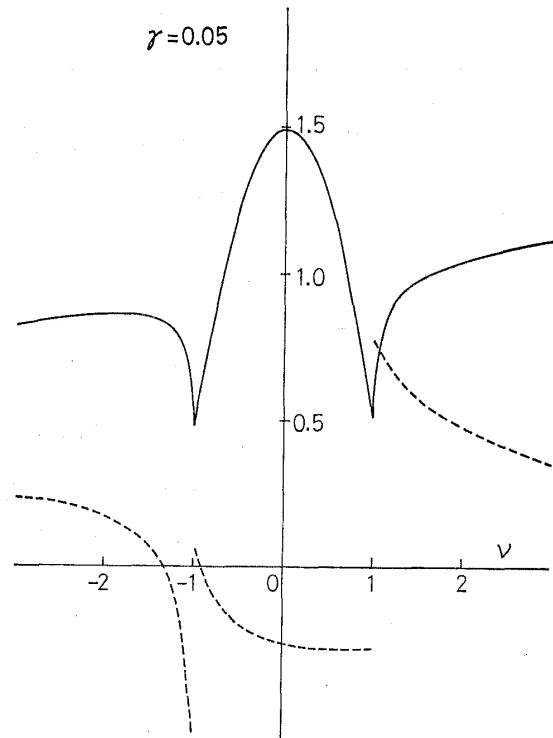


Fig. 9. Zeroth order susceptibility $\chi^{(0)}$ (solid line) (Eq. (4.10)) in unit of χ_K and the correction term 5Φ (dotted lines) due to Coulomb interaction.

§ 5. Summary and discussion

We have developed a gauge- and phase-invariant theory of orbital magnetism of interacting Bloch electrons, including interband effects by use of the Green function method. We first treated the correlated electron system represented by the Hubbard model Hamiltonian in the simplest approximation. Differential susceptibility with respect to a change of number of electrons is shown to have such characteristic features similar to usual Bloch electrons, but with modifications due to interband contributions like Van-Vleck paramagnetism. Vertex corrections, which will play essential roles in some cases, are completely neglected in this approximation. Next, we examined the effect of Coulomb interactions on the simple two-band model of Bloch electrons, where the interband effects are known to be important. The corrections of the order of $r_s \ln r_s$ to free Bloch electrons, which correspond to the discussion by Kanazawa and Matsudaira of the correlation effect on Landau diamagnetism, have a strong dependence on Fermi energy near the zone boundary. In treating this problem we assumed from the first that the periodic potential is a one-dimensional cosine type.

Acknowledgements

The author thanks Professor R. Kubo for helpful guidance and fruitful discussions during his graduate course. He is also grateful to the Sakkokai Foundation for financial support.

Appendix A

If vertex corrections are omitted, there results

$$\begin{aligned} \Omega^{(2)} = & 4\mu_B^2 \sum_{\mu, \nu} A_{q\mu} A_{q\nu} T \sum_n \sum_{\mathbf{k}} \text{Tr} [\gamma_\mu \mathcal{G}(\mathbf{k}^+) \gamma_\nu \mathcal{G}(\mathbf{k}^-) \\ & + \gamma_\nu \mathcal{G}(\mathbf{k}^+) \gamma_\mu \mathcal{G}(\mathbf{k}^-)] + 8m\mu_B^2 A_q^2 T \sum_n \sum_{\mathbf{k}} \text{Tr} \mathcal{G}(\mathbf{k}). \end{aligned} \quad (\text{A} \cdot 1)$$

We expand (A.1) with respect to $\mathbf{q}$. The zeroth order of the first term can be rewritten,

$$\begin{aligned} & \sum_{\mathbf{k}} \text{Tr} [\gamma_\mu \mathcal{G}(\mathbf{k}) \gamma_\nu \mathcal{G}(\mathbf{k}) + \gamma_\nu \mathcal{G}(\mathbf{k}) \gamma_\mu \mathcal{G}(\mathbf{k})] \\ & = \sum_{\mathbf{k}} \text{Tr} m [\gamma_\mu \partial_\nu \mathcal{G}(\mathbf{k}) + \gamma_\nu \partial_\mu \mathcal{G}(\mathbf{k})] \\ & = - \sum_{\mathbf{k}} \text{Tr} m [\partial_\nu \gamma_\mu + \partial_\mu \gamma_\nu] \mathcal{G}(\mathbf{k}) \\ & = -2\delta_{\mu\nu} m \sum_{\mathbf{k}} \text{Tr} \mathcal{G}(\mathbf{k}). \end{aligned} \quad (\text{A} \cdot 2)$$

This term cancels the second of Eq. (A.1). In Eq. (A.2), use is made of the equality

$$\partial_\nu \mathcal{G} = \frac{\partial}{\partial k_\nu} \mathcal{G} = \frac{1}{m} \mathcal{G} \gamma_\nu \mathcal{G} \quad (\text{A} \cdot 3)$$

and the periodicity in momentum space of the quantity γ_μ in applying partial integration. The first order terms vanish. The second order terms can be written as follows by use of the fact $\partial_\mu \gamma_\nu = 0$ for $\mu \neq \nu$ and by partial integrations.

$$4\mu_B^2 \sum_{\mu, \nu, \alpha} A_{q\mu} A_{q\nu} T \sum_n \sum_k [-\delta_{\mu\nu} (1 - \delta_{\mu\alpha}) q_\alpha^2 \gamma_\mu \partial_\alpha \mathcal{G} \gamma_\mu \partial_\alpha \mathcal{G} - \frac{1}{4} \delta_{\mu\nu} \delta_{\mu\alpha} q_\mu^2 (\gamma_\mu \partial_\mu \mathcal{G} \gamma_\mu \partial_\mu \mathcal{G} - \gamma_\mu \partial_{\mu\mu} \mathcal{G} \gamma_\mu \mathcal{G}) - (1 - \delta_{\mu\nu}) q_\mu q_\nu \gamma_\mu \mathcal{G} \gamma_\nu \mathcal{G} \gamma_\mu \mathcal{G} \gamma_\nu \mathcal{G}].$$

Noting that, in general, we can assume $A_z = 0$, $q_z = 0$ for the magnetic field in z -direction and that

$$\begin{aligned} \sum_k \text{Tr} (\gamma_\mu \partial_\mu \mathcal{G} \gamma_\mu \partial_\mu \mathcal{G} - \gamma_\mu \partial_{\mu\mu} \mathcal{G} \gamma_\mu \mathcal{G}) \\ = \frac{2}{3} \sum_k \text{Tr} \partial_\mu (\mathcal{G} \gamma_\mu)^3 = 0 \end{aligned}$$

by partial integration, we get Eq. (2.14).

Appendix B

We prove the equivalence of Eq. (2.15b) to the exact formula by comparing it with the intermediate stage of the derivation in dielectric formulation given by Ichimaru.¹⁵⁾

First, we note that the following equation holds in a matrix form with respect to band indices

$$\frac{1}{m} G j_y G = \partial_y G + i(GY - YG), \quad (\text{B.1})$$

where $\partial_y G \equiv (\partial/\partial k_y) G$ and Y is defined in Eq. (7) of reference 15) [hereafter referred to as, e.g., R-(7)]. Then

$$\begin{aligned} \frac{4\mu_B^2}{m^2} \text{Tr} G j_y G j_x G j_y G j_x \\ = 4\mu_B^2 \text{Tr} [\partial_y G + i(GY - YG)] j_x [\partial_y G + i(GY - YG)] j_x \\ = 4\mu_B^2 \text{Tr} \partial_y G j_x \partial_y G j_x + 8i\mu_B^2 \text{Tr} [\partial_y G j_x G Y j_x - G j_x \partial_y G j_x Y] \\ + 8\mu_B^2 \text{Tr} G Y j_x Y G j_x - 4\mu_B^2 \text{Tr} G Y j_x G Y j_x - 4\mu_B^2 \text{Tr} G j_x Y G j_x Y. \end{aligned} \quad (\text{B.2})$$

Intraband contributions defined in reference 15) correspond to terms that have the same band indices in the two Green functions in (B.2), and are classified as follows. Summation over thermal frequency of the first term in (B.2) yields

$$\frac{2}{3} m^2 \mu_B^2 \sum_{l, k} \left(\frac{\partial \epsilon_l}{\partial k_x} \right)^2 \left(\frac{\partial \epsilon_l}{\partial k_y} \right)^2 \frac{\partial^3 f_l}{\partial \epsilon_l^3} \quad (f_l \equiv f(\epsilon_l))$$

which is equal to the second term of R-(36). The term in the curly bracket in (B.2) gives, with the aid of (2.20),

$$4m^2 \mu_B^2 \frac{\partial \epsilon_l}{\partial k_x} \frac{\partial \epsilon_l}{\partial k_y} \frac{\partial^2 \epsilon_l}{\partial k_x \partial k_y} \frac{\partial^2 f_l}{\partial \epsilon_l^2}. \quad (\text{B.3})$$

The third term of (B·2) is

$$8m\mu_B^2 \sum_{l,k} \frac{\partial \varepsilon_l}{\partial k_x} (Yj_x Y)_u \frac{\partial f_l}{\partial \varepsilon_l}$$

which is the same as the last term of R-(36). The last two terms of (B·2) are

$$-4\mu_B^2 \sum_{l,k} \frac{\partial f_l}{\partial \varepsilon_l} [(Yj_x)_u^2 + (j_x Y)_u^2]$$

which, by use of the relation (W_l are defined by R-(7))

$$(Yj_x)_u = W_l - \frac{i}{2} m \frac{\partial^2 \varepsilon_l}{\partial k_x \partial k_y},$$

$$(j_x Y)_u = W_l + \frac{i}{2} m \frac{\partial^2 \varepsilon_l}{\partial k_x \partial k_y}$$

yields

$$-8\mu_B^2 \sum_{l,k} \frac{\partial f_l}{\partial \varepsilon_l} W_l^2 + 2m^2 \mu_B^2 \sum_{l,k} \frac{\partial f_l}{\partial \varepsilon_l} \left(\frac{\partial^2 \varepsilon_l}{\partial k_x \partial k_y} \right)^2. \quad (\text{B} \cdot 4)$$

The first of (B·4) is identical to the fourth of R-(36). The second of (B·4) and (B·3) are added to be equal to the sum of the first and third term of R-(36), as the latter can be transformed as follows by partial integration:

$$\begin{aligned} m^2 \mu_B^2 \sum_{l,k} \frac{\partial}{\partial k_y} \left(\frac{\partial \varepsilon_l}{\partial k_x} \right)^2 \cdot \frac{\partial \varepsilon_l}{\partial k_y} \frac{\partial^2 f_l}{\partial \varepsilon_l^2} + 2m^2 \mu_B^2 \sum_{l,k} \frac{\partial}{\partial k_y} \left(\frac{\partial \varepsilon_l}{\partial k_x} \frac{\partial f_l}{\partial \varepsilon_l} \right) \frac{\partial^2 \varepsilon_l}{\partial k_x \partial k_y} \\ = 4m^2 \mu_B^2 \sum_{l,k} \frac{\partial \varepsilon_l}{\partial k_x} \frac{\partial \varepsilon_l}{\partial k_y} \frac{\partial^2 \varepsilon_l}{\partial k_x \partial k_y} \frac{\partial^2 f_l}{\partial \varepsilon_l^2} + 2m^2 \mu_B^2 \sum_{l,k} \left(\frac{\partial^2 \varepsilon_l}{\partial k_x \partial k_y} \right)^2 \frac{\partial f_l}{\partial \varepsilon_l}. \end{aligned} \quad (\text{B} \cdot 5)$$

We thus proved the equivalence of intraband contributions.

Next, in order to discuss interband terms, we further transform the second line of (B·2) by partial integration as follows (factor $4\mu_B^2$ is omitted for the sake of convenience)

$$\begin{aligned} -\text{Tr } \partial_y [j_x \{ \partial_y G + i(GY - YG) \} j_x] G + i \text{Tr } Yj_x [\partial_y G \\ + i(GY - YG)] j_x G - i \text{Tr } j_x [\partial_y G + i(GY - YG)] j_x YG. \end{aligned} \quad (\text{B} \cdot 6)$$

Performing a differentiation and using the relation

$$\partial_y j_x = i(Yj_x - j_x Y), \quad (\text{B} \cdot 7)$$

we get

$$\begin{aligned} -\text{Tr } \partial_y^2 G j_x G j_x + 2i \text{Tr } (j_x Y \partial_y G j_x G - j_x \partial_y G Y j_x G) - 2 \text{Tr } j_x Y G Y j_x G \\ + \text{Tr } j_x (Y^2 + i \partial_y Y) G j_x G + \text{Tr } (Y^2 - i \partial_y Y) j_x G j_x G. \end{aligned} \quad (\text{B} \cdot 8)$$

Summation over thermal frequency in the first term (B·8) yields ($j_{lu}^\mu \equiv (j_\mu)_u$)

$$\begin{aligned}
 & - \sum'_{i,l',k} \left[2 \left(\frac{\partial \varepsilon_i}{\partial k_y} \right)^2 j_{il'}^x j_{il'}^x \left(\frac{1}{\varepsilon_i - \varepsilon_{l'}} \frac{\partial^2 f_i}{\partial \varepsilon_i^2} - 2 \frac{1}{(\varepsilon_i - \varepsilon_{l'})^2} \frac{\partial f_i}{\partial \varepsilon_i} + 2 \frac{f_i}{(\varepsilon_i - \varepsilon_{l'})^3} \right) \right. \\
 & \quad \left. - \frac{\partial^2 \varepsilon_i}{\partial k_y^2} j_{il'}^x j_{il'}^x \left(\frac{f_i}{(\varepsilon_i - \varepsilon_{l'})^2} - \frac{1}{(\varepsilon_i - \varepsilon_{l'})} \frac{\partial f_i}{\partial \varepsilon_i} \right) \right] \\
 & - \sum'_{i,l',k} \frac{j_{il'}^x j_{il'}^x}{(\varepsilon_i - \varepsilon_{l'})^2} f_{l'} \left[2 \left(\frac{\partial \varepsilon_i}{\partial k_y} \right)^2 \frac{1}{(\varepsilon_{l'} - \varepsilon_i)} + \frac{\partial^2 \varepsilon_i}{\partial k_y^2} \right]. \quad (B.9)
 \end{aligned}$$

The first curly bracket of (B.9) with a remaining factor added gives half of the last four terms of χ^{inter} on p. 528 of reference 15) (hereafter referred to as χ^{inter}), which stem from the q^2 -terms in R-(43). The second of (B.8) is

$$\begin{aligned}
 & 2i \sum'_{i,l',k} \frac{\partial \varepsilon_i}{\partial k_y} [(j_x Y)_{il'} j_{il'}^x - (Y j_x)_{il'} j_{il'}^x] \left[\frac{1}{\varepsilon_i - \varepsilon_{l'}} \frac{\partial f_i}{\partial \varepsilon_i} - \frac{1}{(\varepsilon_i - \varepsilon_{l'})^2} f_i \right] \\
 & + 2i \sum'_{i,l',k} \frac{\partial \varepsilon_i}{\partial k_y} \frac{f_{l'}}{(\varepsilon_i - \varepsilon_{l'})^2} [(j_x Y)_{il'} j_{il'}^x - (Y j_x)_{il'} j_{il'}^x]. \quad (B.10)
 \end{aligned}$$

The first term of (B.10) makes half of the third term of χ^{inter} , which arises from the product of q -linear terms in R-(43) and R-(44). The third of (B.8) becomes

$$-2 \sum'_{i,l',k} \frac{f_i}{\varepsilon_i - \varepsilon_{l'}} |(j_x Y)_{il'}|^2 + 2 \sum'_{i,l',k} \frac{f_{l'}}{\varepsilon_i - \varepsilon_{l'}} |(j_x Y)_{il'}|^2 \quad (B.11)$$

the first of which is equal to the half of the first term of χ^{inter} . Similarly the last two terms of (B.8) can be written as

$$\begin{aligned}
 & \sum'_{i,l',k} j_{il'}^x [\{j_x (Y^2 + i\partial_y Y)\}_{il'} + \{(Y^2 - i\partial_y Y) j_x\}_{il'}] \frac{f_i}{\varepsilon_i - \varepsilon_{l'}} \\
 & - \sum'_{i,l',k} j_{il'}^x [\{j_x (Y^2 + i\partial_y Y)\}_{il'} + \{(Y^2 - i\partial_y Y) j_x\}_{il'}] \frac{f_{l'}}{\varepsilon_i - \varepsilon_{l'}}. \quad (B.12)
 \end{aligned}$$

The first term of (B.12) yields half of the second term of χ^{inter} . Remaining terms which we did not identify in (B.9), (B.10), (B.11) and (B.12), and which include $f_{l'}$ and its derivative, are easily seen to be q^2 -terms in

$$2 \sum_{\substack{k \\ l \neq l'}} \frac{f(\varepsilon_{l'k})}{\varepsilon_{lk+q} - \varepsilon_{l'k}} |(l'k|p_x + k_x|lk+q)|^2 \quad (B.13)$$

but (B.11) is equal to

$$-2 \sum_{\substack{k \\ l \neq l'}} \frac{f(\varepsilon_{lk+q})}{\varepsilon_{lk+q} - \varepsilon_{l'k}} |(l'k|p_x + k_x|lk+q)|^2 \quad (B.14)$$

by time reversal symmetry whereas (B.14) constitutes half of χ^{inter} . Thus we proved the required equivalence.

Appendix C

Perturbational treatment of the periodic potential V for orbital susceptibility are given by Glasser,²⁶⁾ Samoilovich and Rabinovich,²⁷⁾ and Yamaji and Kubo²⁸⁾

(Y.K.). The last authors obtained for the bcc crystal the exact coefficient in the second order of the periodic potential; and at the same time they evaluated the value due to the Landau-Peierls approximation. We now calculate the value by $\bar{\chi}$, Eq. (2.18) and χ_{HS} Eq. (2.19) for a comparison of the one-band formula. To the second order of V , the band energy is given by (Eq. (15) of Y.K.)

$$E(\mathbf{k}) = E_0(\mathbf{k}) + \sum_{\mathbf{K}}' \frac{|V_{\mathbf{K}}|^2}{E_0(\mathbf{k}) - E_0(\mathbf{k} + \mathbf{K})}.$$

Then $(\partial^2/\partial k_x \partial k_y) E(\mathbf{k}) = 0(V^2)$, therefore $\bar{\chi}$ and χ_{HS} coincide to the second order of V . Using

$$\frac{\partial^3 E(\mathbf{k})}{\partial k_y \partial k_x^2} = 96m \sum_{\mathbf{K}}' \frac{V_{\mathbf{K}}^2}{(\mathbf{K}^2 + 2\mathbf{K} \cdot \mathbf{k})^4} K_x^2 K_y,$$

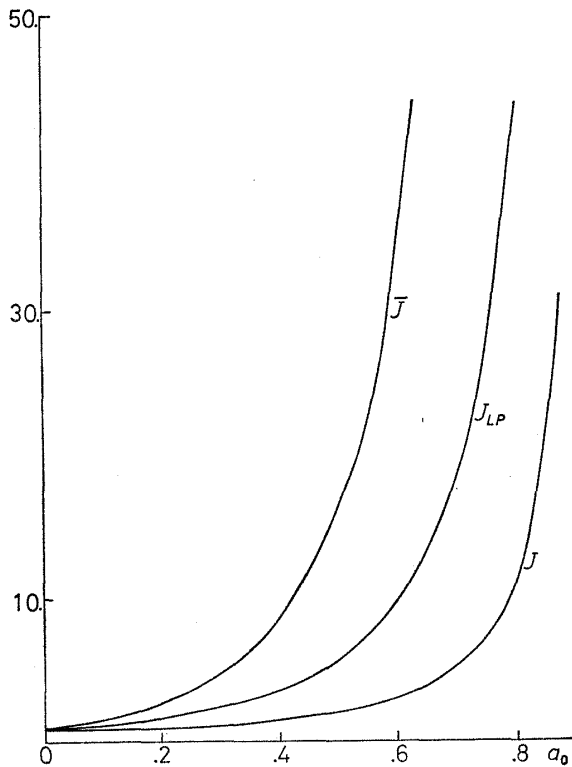
we have for bcc metal (notations are the same as in Y.K.)

$$\bar{\chi} - \chi_{\text{LP}} = -\frac{8}{3} \chi_{\text{L}} \left| \frac{V_{\mathbf{K}}}{E_0(\mathbf{K}/2)} \right|^2 \cdot \frac{a_0}{(1-a_0)^3}$$

and then

$$\bar{\chi} = \chi_{\text{L}} \left\{ 1 - \left| \frac{V_{\mathbf{K}}}{E_0(\mathbf{K}/2)} \right|^2 \bar{J}(a_0) \right\},$$

$$\bar{J}(a_0) = \frac{3}{4} a_0^{-3/2} \ln \left| \frac{1 + \sqrt{a_0}}{1 - \sqrt{a_0}} \right| - \frac{3-7a_0}{2a_0(1-a_0)^2} + \frac{8}{3} \frac{a_0}{(1-a_0)^3}.$$



$\bar{J}(a_0)$ is shown in Fig. 10 with J_{orb} and J_{LP} of Y.K. From this, $\bar{\chi}$ and χ_{HS} are less valid than χ_{LP} , at least in the case of a weak periodic potential, but that which is the best near the zone boundary with not too small a band gap is open to question.

Fig. 10. The coefficients of the susceptibility in the second order of periodic potential for bcc crystals by exact calculation J , Landau-Peierls approximation J_{LP} and Eq. (2.18) or (2.19) $\bar{J}$. J and J_{LP} are due to reference 28).

Appendix D

Equation (3.26) can be transformed as follows:

$$\begin{aligned}\bar{\chi} &= -\frac{2m^2\mu_B^2}{3\pi} \frac{\partial^2}{\partial\mu^2} \sum_{\pm} \delta(\mu - E_{\pm}) Z^4(\mathbf{k}) \left(\frac{\partial\varepsilon}{\partial k_x}\right)^2 \left(\frac{\partial\varepsilon}{\partial k_y}\right)^2 \\ &= -\frac{2m^2\mu_B^2}{3\pi} \frac{\partial^2}{\partial\mu^2} z'(\mu)^{-4} \sum_{\pm} z'(\mu) \delta(\nu - \varepsilon(\mathbf{k})) \left(\frac{\partial\varepsilon}{\partial k_x}\right)^2 \left(\frac{\partial\varepsilon}{\partial k_y}\right)^2 \\ &= -\frac{m^2\mu_B^2}{45\pi^2 V_0} v^4 \frac{\partial^2}{\partial\mu^2} \frac{1}{z'(\mu)^3} [1 - z(\mu)^2]^{5/2}.\end{aligned}$$

This gives Eq. (3.28).

Appendix E

We derive Eq. (4.23). First let us consider $C_q^{(2)}$.

$$\begin{aligned}C_q^{(2)} &= \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} f_{\pm}(\mathbf{k}) f_{\pm}^{(2)}(\mathbf{k}+\mathbf{q}) \\ &= \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} f_{\pm}(\mathbf{k}) \frac{\partial^2}{\partial\mu^2} f_{\pm}(\mathbf{k}+\mathbf{q}) \\ &= \frac{\partial}{\partial\mu} \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} f_{\pm}(\mathbf{k}) \frac{\partial}{\partial\mu} f_{\pm}(\mathbf{k}+\mathbf{q}) - \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} \frac{\partial}{\partial\mu} f_{\pm}(\mathbf{k}) \frac{\partial}{\partial\mu} f_{\pm}(\mathbf{k}+\mathbf{q}).\end{aligned}\tag{E.1}$$

The first term of Eq. (E.1) does not contain $1/q$, whereas the second term can be transformed for small q as follows:

$$\begin{aligned}& - \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} \delta(\mu - E_{\pm}(\mathbf{k})) \delta(\mu - E_{\pm}(\mathbf{k}+\mathbf{q})) \\ & \cong - \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} \delta(\mu - E_{\pm}(\mathbf{k})) \delta(\mu - E_{\pm}(\mathbf{k}) - \mathbf{q} \cdot \nabla_{\mathbf{k}} E_{\pm}(\mathbf{k})) \\ & = - \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} \delta(\mu - E_{\pm}(\mathbf{k})) \delta(\mathbf{q} \cdot \nabla_{\mathbf{k}} E_{\pm}(\mathbf{k})) \\ & \cong - \frac{1}{q} \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}} \delta(\mu - E_{\pm}(\mathbf{k})) \delta(\mathbf{e} \cdot \nabla_{\mathbf{k}} E_{\pm}(\mathbf{k}))\end{aligned}\tag{E.2}$$

for small q where $\mathbf{e}$ is the unit vector of the direction of q . Similarly

$$\begin{aligned}C_q^{(3)} &= \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} f_{\pm}(\mathbf{k}) f_{\pm}^{(3)}(\mathbf{k}+\mathbf{q}) = - \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} f_{\pm}(\mathbf{k}) \frac{\partial^3 f_{\pm}(\mathbf{k}+\mathbf{q})}{\partial\mu^3} \\ &= - \frac{\partial}{\partial\mu} \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} f_{\pm} \frac{\partial^2}{\partial\mu^2} f_{\pm}(\mathbf{k}+\mathbf{q}) + \sum_{\mathbf{k}} \zeta_{\mathbf{k}} \eta_{\mathbf{k}+\mathbf{q}} \frac{\partial}{\partial\mu} f_{\pm}(\mathbf{k}) \frac{\partial^2}{\partial\mu^2} f_{\pm}(\mathbf{k}+\mathbf{q}).\end{aligned}\tag{E.3}$$

By (E.1) and (E.2), the first term of (E.3) yields

$$\frac{1}{q} \sum_{\mathbf{k}} \frac{\partial}{\partial\mu} \delta(\mu - E_{\pm}(\mathbf{k})) \delta(\mathbf{e} \cdot \nabla_{\mathbf{k}} E_{\pm}(\mathbf{k})) \zeta_{\mathbf{k}} \eta_{\mathbf{k}}.$$

If $\zeta_{\mathbf{k}}$ is independent of $\mathbf{k}$, integration over $\mathbf{k}$ can be performed in the second term of (E.3) and this is found to give same contribution as the first one as regards the singularity. This is due to the relation

$$\frac{\partial}{\partial \mu} \sum_q \frac{1}{\kappa^2 + q^2} \delta(\mu - E_{\pm}(\mathbf{k} + \mathbf{q})) = 0$$

as far as the singularity $\ln \kappa^2$ is concerned. Thus we get, omitting terms that do not include the above-mentioned singularity,

$$\begin{aligned} \chi^{(2)} = & \mu_B \frac{e^2}{\pi} \ln \kappa^2 \sum_{\pm} \left[\left(1 \pm \frac{\lambda^2 V^2}{X^3} \right) \frac{\delta(\mu - E_{\pm})}{|\nabla_{\mathbf{k}} E_{\pm}|} \right. \\ & \left. - \frac{2}{3} \left(\pm \frac{k_x^2 \lambda^2 V^2}{X^3} + k_x^2 + j_{\pm}^2 \right) \frac{1}{|\nabla_{\mathbf{k}} E_{\pm}|} \frac{\partial}{\partial \mu} \delta(\mu - E_{\pm}) \right]. \end{aligned}$$

Integrating over k_x and k_z , we get Φ in Eq. (4.23).

References

- 1) L. Landau, Z. Phys. **64** (1930), 629.
- 2) R. Peierls, Z. Phys. **80** (1933), 763.
- 3) N. H. March and B. Donovan, Proc. Phys. Soc. **A67** (1964), 464.
H. Kanazawa, Prog. Theor. Phys. **15** (1956), 273; **17** (1957), 1.
J. G. Fletcher and D. C. Larson, Phys. Rev. **111** (1958), 455.
G. Wentzel, Phys. Rev. **108** (1957), 1593.
- 4) H. Kanazawa and N. Matsudaira, Prog. Theor. Phys. **23** (1960), 433.
- 5) S. Tani, Prog. Theor. Phys. **23** (1960), 1157.
- 6) H. Shiba, H. Fukuyama and K. Yamaji, Reported in the Autumnal Meeting of Physical Society of Japan (1967).
- 7) H. Schmidt, Z. Phys. **216** (1968), 336.
A. Schmid, Phys. Rev. **180** (1969), 527.
K. Yamaji, Phys. Letters **29A** (1969), 123.
M. Koyanagi and T. Tsuzuki, Phys. Letters **30A** (1969), 545.
- 8) R. B. Dingle, Proc. Roy. Soc. **A211** (1952), 517.
- 9) M. Saitoh, H. Fukuyama, Y. Uemura and H. Shiba, J. Phys. Soc. Japan **27** (1969), 26.
- 10) D. Shönberg and M. Z. Uddin, Proc. Roy. Soc. **A156** (1936), 687.
- 11) A. H. Wilson, Proc. Cambridge Phil. Soc. **49** (1953), 292.
- 12) E. N. Adams, Phys. Rev. **89** (1953), 633.
- 13) J. E. Hebborn and E. H. Sondheimer, Phys. Rev. Letters **2** (1959), 150; J. Phys. Chem. Solids **13** (1960), 105.
- 14) J. E. Hebborn, J. M. Luttinger, E. H. Sondheimer and P. L. Stiles, J. Phys. Chem. Solids **23** (1962), 433.
E. I. Blount, Phys. Rev. **126** (1962), 1636.
L. M. Roth, J. Phys. Chem. Solids **23** (1962), 443; Phys. Rev. **145** (1966), 434.
G. H. Wannier and U. N. Upadhyaya, Phys. Rev. **136** (1964), A803.
- 15) S. Ichimaru, J. Phys. Soc. Japan **20** (1965), 520.
- 16) H. Fukuyama and R. Kubo, J. Phys. Soc. Japan **28** (1970), 570.
- 17) J. Hubbard, Proc. Roy. Soc. **A276** (1963), 238.
- 18) H. Fukuyama and R. Kubo, J. Phys. Soc. Japan **27** (1969), 604.
- 19) J. Ruvalds, J. Phys. Chem. Solids **30** (1969), 305.

- 20) J. M. Luttinger and W. Kohn, Phys. Rev. **97** (1955), 869.
- 21) H. Fukuyama, Prog. Theor. Phys. **42** (1969), 1284.
- 22) A. H. Wilson, *The Theory of Metals* (Camb. Univ. Press, 1953), Chap. 2.
- 23) A. Sakurai, Prog. Theor. Phys. **30** (1968), 312.
- 24) H. Fukuyama, Prog. Theor. Phys. **44** (1970), 879.
- 25) J. H. Van-Vleck, *The Theory of Electric and Magnetic Susceptibility* (Clarendon Press, Oxford, 1932), §§ 44, 50, 52.
- 26) M. L. Glasser, Phys. Rev. **134** (1964), A1296.
- 27) A. G. Samoilovich and E. Ya. Rabinovich, Soviet Phys.—Solid State **5** (1963), 567.
- 28) K. Yamaji and R. Kubo, J. Phys. Soc. Japan **25** (1968), 330.